

LServe + SampleAttention Slide Deck

Review online here, or download the prebuilt PDF for a LinkedIn document upload.

PDF: [./lservice-sampleattention.pdf](#)

Post URL: [../blog/2026-01-30-lservice-sampleattention-sparse-attention/](#)

LONG-CONTEXT INFERENCE

Sparse Attention That Actually Changes Serving

LServe explains the serving path. SampleAttention explains the sparse structure.

SHARED THESIS

**The real win comes from structured sparsity
that kernels can exploit cheaply.**

LServe

Serving-time view

Use different sparsity
for different bottlenecks:

Prefill skips KV blocks.
Decode prunes KV pages.



Block-sparse prefill



Page-pruned decode

SampleAttention

Sparse-structure view

The key claim

Two recurring patterns explain
much of the useful attention mass:
columns for global anchors, slashes for local continuations



CRA acts like a runtime accuracy guardrail.

LServe: Two Phases, Two Sparse Mechanisms

The paper is most useful when you stop treating prefill and decode as the same optimization

Prefill

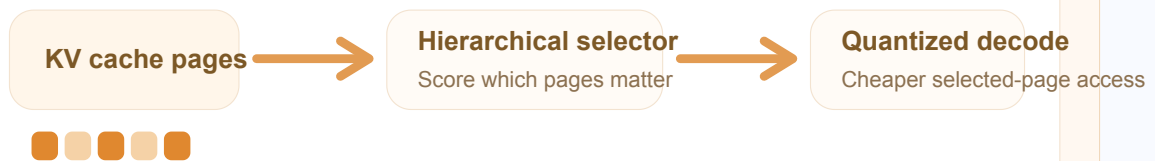
Goal: reduce sequential KV-block work over many query tokens.



Mental model: prefill acceleration comes from skipping whole blocks, not a few scalar ops inside each block.

Decode

Goal: reduce memory-bound KV-page reads for one query token at a time.



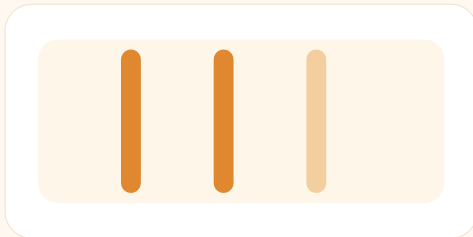
Mental model: prefill skips blocks; decode prunes pages, then makes each selected page cheaper to read.

SampleAttention: The Sparse Patterns That Matter Most

Its key contribution is not just sparsity. It is runtime-adaptive sparsity with an accuracy proxy

Column pattern

Global anchors and attention sinks



Many queries still need the same important key position.
That is why a few columns can carry a lot of useful mass.

Slash pattern

Local continuity and recent context



Queries often attend along relative-position bands.
That is why slash-like structure preserves local behavior.

CRA = Cumulative Residual Attention

Defined as the minimum remaining attention probability mass per query after sparsification.



Why it matters: CRA is a runtime proxy for whether the sparse mask kept enough useful attention.
SampleAttention approximates it with chunked query sampling, then picks top-k column and slash blocks.

What I Would Keep from Both Papers

A production-focused mental model for long-context inference.

1

Prefill and decode are different optimization problems.

If the bottlenecks differ, the sparse mechanism should differ too.

2

The best sparse patterns are structured, not arbitrary.

Columns preserve global anchors. Slashes preserve local continuity.

3

Accuracy needs a runtime control signal.

CRA is better than a fixed sparsity ratio or offline pattern choice.

Bottom line

SampleAttention tells you what to keep during prefill.

LServe tells you how to serve sparse attention across prefill and decode.